

CMSC6950 Homework #2

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Question 1

Q1a. Assign values to two variables that you cannot add. Show the error message from trying to add them.

```
In [1]: country = "Canada"
        n_province = 10

        print(country + n_province)
```

```
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TypeError                                Traceback (most recent call last)
Cell In[1], line 4
      1 country = "Canada"
      2 n_province = 10
----> 4 print(country + n_province)

TypeError: can only concatenate str (not "int") to str
```

Q1b. Define a string variable for a sentence of at least 6 words. Use slicing to print only the third and fourth words. Use slicing to print only the last word.

```
In [9]: task = "Perform each of the following tasks in a single Jupyter notebook"

        words = task.split()
        print(words)

        third_and_fourth_words = words[2:4]
        print("Third and fourth words:", third_and_fourth_words)

        last_word = words[-1]
        print("Last word:", last_word)
```

```
['Perform', 'each', 'of', 'the', 'following', 'tasks', 'in', 'a', 'single', 'Jupyter', 'notebook']
Third and fourth words: ['of', 'the']
Last word: notebook
```

Q1c. We define $\text{plex}(n)$ to be a 1 followed by n zeros, so that $\text{plex}(2)$ is 100, and $\text{plex}(6)$ is 1 million. Write a python function that takes an integer, n as input, with

default value 0, prints a statement such as The plex of 3 is 1000. (correctly substituting the input value, n, and plex(n)) and returns the value of plex(n). Run your code for 3 sample values of n, at least one of which is a single digit, and at least one which is greater than 20.

```
In [ ]: def plex(n=0):  
        """  
        Input:  
            n: Integer (Default = 0)  
  
        Returns:  
            Plex of a given integer, 10^n  
        """  
  
        val = 10**n  
        print(f"The plex of {n} is {val}")  
        return val  
  
plex()
```

The plex of 0 is 1

```
Out[ ]: 1
```

```
In [18]: sample = (7, 15, 24)  
  
for s in sample:  
    plex(s)
```

The plex of 7 is 10000000

The plex of 15 is 1000000000000000

The plex of 24 is 10000000000000000000000000

Question 2

```

Last login: Mon Sep 29 16:53:57 on ttys006

isaacadoboe@mac ~
(base) > $ Developer/scientific-comp/HW2 [17:13:18]

isaacadoboe@mac ~/Developer/scientific-comp/HW2 [17:13:41]
(base) > $ mkdir Isaac_CMSC6950 [main ●]

isaacadoboe@mac ~/Developer/scientific-comp/HW2 [17:14:09]
(base) > $ cd Isaac_CMSC6950 [main ●]

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:14:25]
(base) > $ git init [main ●]
Initialized empty Git repository in /Users/isaacadoboe/Developer/scientific-comp/HW2/Isaac_CMSC6950/.git/

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:14:28]
(base) > $ nano HW2.txt [main ↵]

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:15:23]
(base) > $ git status [main ●]
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store
        HW2.txt

nothing added to commit but untracked files present (use "git add" to track)

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:15:38]
(base) > $ git add HW2.txt [main ●]
warning: in the working copy of 'HW2.txt', LF will be replaced by CRLF the next time Git touches it

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:16:10]
(base) > $ git commit -m"initial commit, passenger flight info" [main ●]
[main (root-commit) f7e53e6] initial commit, passenger flight info
1 file changed, 3 insertions(+)
create mode 100644 HW2.txt

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:16:43]
(base) > $ git status [main ●]
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store

nothing added to commit but untracked files present (use "git add" to track)

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:18:24]
(base) > $ nano HW2.txt [main ●]

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:18:53]
(base) > $ git diff [main ●]

```

```

warning: in the working copy of 'HW2.txt', LF will be replaced by CRLF the next time Git touches it
diff --git a/HW2.txt b/HW2.txt
index d773501..adc4207 100644
--- a/HW2.txt
+++ b/HW2.txt
@@ -1,3 +1,2 @@
   Isaac Adoboe
-Flight 123
-Toronto to Halifax
(END)

```

```

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:19:25]
(base) > $ git add HW2.txt [±main ●●]
warning: in the working copy of 'HW2.txt', LF will be replaced by CRLF the next time Git touches it

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:19:51]
(base) > $ git commit -m"removed flight number" [±main ●●]
[main 911d6d8] removed flight number
1 file changed, 1 deletion(-)

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:20:04]
(base) > $ nano .gitignore [±main ●]

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:21:57]
(base) > $ git status [±main ●]
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:22:10]
(base) > $ git add .gitignore [±main ●]
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:22:20]
(base) > $ git commit -m"added gitignore to root" [±main ●]
[main 8548799] added gitignore to root
1 file changed, 7 insertions(+)
create mode 100644 .gitignore

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:23:17]
(base) > $ touch important.dat runtime.log inputs.dat [±main ↗]

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:24:31]
(base) > $ git add important.dat runtime.log inputs.dat [±main ●]
The following paths are ignored by one of your .gitignore files:
inputs.dat
runtime.log
hint: Use -f if you really want to add them.
hint: Disable this message with "git config set advice.addIgnoredFile false"

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:25:16]
(base) > $ git commit -m"added important data file" [±main ●]
[main fc2dbcc] added important data file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 important.dat

isaacadoboe@mac ~/Developer/scientific-comp/HW2/Isaac_CMSC6950 [17:26:18]
(base) > $ git log --oneline [±main ↗]

```

```

fc2dbcc (HEAD -> main) added important data file
8548799 added gitignore to root
911d6d8 removed flight number
f7e53e6 initial commit, passenger flight info
(END)

```